Data Mining, People and Society

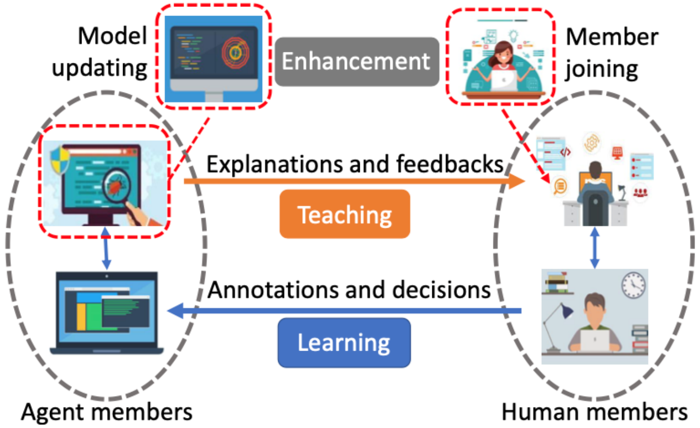
- Privacy-Preserving Data Mining
- Human-Algorithm Interaction
- Mining beyond Maximizing Accuracy: Fairness, Interpretability, and Robustness
- Data Mining for Social Good

Human-Algorithm Interaction

- Human-algorithm interaction
  - Goal: optimize the collaboration between human intelligence and algorithm, and consequently enhance the system performance and user experience.
- Crowdsourcing
  - Goal: harness the human intelligence to resolve difficult problems.
  - Applications: label generation, model evaluation, evaluating prediction interpretability, debug AI systems, etc.
- Human-in-the-loop
  - Goal: leveraging human intervention for improving algorithms.
  - Applications: high-quality label generation for supervised learning, enhancing model explainability with background knowledge, model evaluation, etc.

Human-Algorithm Interaction

- Machine-in-the-loop
  - Goal: improving humans' query strategy by algorithms.
  - Applications: creative writing and drawing, achieving comparable model performance which reducing the demand for labelled data
- Human-machine-teaming
  - Goal: fostering the effective teamwork between humans and algorithms.
  - Applications:



(a) HMT Overview

(b) HMT for Cyber Defense